Ashen Gunaratne

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experience

Secure Agility — Sydney, Australia
Lead DevOps Engineer — August 2023 - Present

Lead cross-SaaS integration architect; Design, author, and oversea custom integrations and workflows interconnecting multiple SaaS services.

Alembic · Autotask · Azure DevOps Services · Azure · Ekahuau · Float · GitHub · ITGlue · LogicMonitor · PowerBI · Python · Terraform

Singlife — Singapore, Singapore
Technical Lead — April 2022 - July 2023

Mobile backend tech lead; Design, author, and review RESTful API specifications and moderate related infrastructure cohering to compliance and security statues.

Anti-Corruption Layer · AWS · AWS CDK · Jest · Node.js · Quantum Databases · RESTful APIs · Slack Bots · TeamCity · TypeScript

Railsbank — London, United Kingdom
Associate Technical Lead — December 2021 - March 2022

BaaS integrations tech lead; Plan, design, and oversea regulated provider infrastructure integrations conforming to compliance and security statutes.

AWS · CucumberJS · Elastic Stack (ELK) · GitLab CI/CD · GraphQL · Jest · Kubernetes · mTLS · Node.js · OpenPGP · RESTful APIs · Slack Bots · Terraform · TypeScript

Railsbank — London, United Kingdom
Engineer — December 2020 - December 2021

BaaS integrations engineer; Design, author, and review regulated provider infrastructure integrations conforming to compliance and security statutes.

AWS · CucumberJS · Elastic Stack (ELK) · GitLab CI/CD · GraphQL · Jest · Kubernetes · mTLS · Node.js · OpenPGP · RESTful APIs · Slack Bots · Terraform · TypeScript

Al Iman Schools — Colombo, Sri Lanka
Educator — October 2020 - December 2020

Educator for computer science; Responsible for designing, implementing, and delivering custom tailored curriculums for key stages three, four, and five with emphasis on art of programming and integrating computer science in the domain of other subjects.

AsciiDoc · CSS · HTML · JavaScript · Python · PyUnit

British College of Applied Studies — Colombo, Sri Lanka
Lecturer — July 2018 - July 2019

Instructor for *Pearson BTEC HND* in Computing; Responsible for level 5 specialist units: *Computer Systems Architecture*, *Data Structures and Algorithms*, *Discrete Mathematics*, *Network Management*, and *Network Security*.

C · Firewalls · IDS/IPS Systems · JavaScript · Python

Exetel PTY Limited — Sydney, Australia
Network Engineer — October 2016 - February 2017

Govern, implement, and troubleshoot real-time traffic steering policies to ensure critical application delivery, value-added services, and user quality of experience driving profitability of key deliverables in the digital data & device space in terms of revenue, users, traffic across 30 Gb/s data link.

Firewalls IPv4 RADIUS Servers SNMP TCP/UPD VoIP

IKON Systems (Private) Limited — Colombo, Sri Lanka
Chief Information Officer — January 2016 - October 2016

Direct and manage computing and information technology strategic plans, policies, programs, schedules, and responsible for advertising network's scalability, security, and capacity-planning.

AWS Systemd HAProxy

open source contributions

Azure Resource Manager
Terraform Provider

Contributions include feature implementations and bug fixes on existing resource types and data sources.

Databricks
Terraform Provider

Contributions include feature implementations and bug fixes on existing resource types and data sources.

research interests

Cybersecurity, digital forensics, botnets, computer science education, distance learning, collaborative learning, computer-assisted instruction

education

Harvard College — Massachusetts, United States
Bachelor of Liberal Arts (A.L.B.) — June 2017 - September 2021

Computer Science; GPA 3.9 (as of May 2018)
Related studies in linear algebra, statistics and probability, and finance.

Stanford University — California, United States
MOOC — October 2017 - November 2017

Studies in enterprise-level web application deployment with focus on security, reliability, scalability, and cross-browser support.

Harvard University — Massachusetts, United States
MOOC — January 2015 - December 2015

Studies in computer science for real-world domains of biology, cryptography, finance, forensics, and gaming.

Université Catholique de Louvain — Charleroi, Belgium
MOOC — January 2015 - November 2015

Studies in major programming concepts, techniques, and paradigms in a unified framework.

The University of British Columbia — Vancouver, Canada
MOOC — January 2015 - October 2015

Studies in computation as a tool for systematic problem solving in non-computer-science disciplines.

Lakeland Inter-American School — Colombo, Sri Lanka
Pearson Edexcel AS Level — June 2014 - July 2015

Lakeland Inter-American School — Colombo, Sri Lanka
Pearson Edexcel Ordinary Level — March 2011 - June 2014

projects

Singlife's BaaS provider reverse-engeering

- Associate technical lead for Railsbank replacement with in-house BaaS service
- Coarchitectured in-house replacement service to extactly replicate Railsbank functionalities
- Designed and authored the state machines for transaction execution - interledger, manual credit, and manual debit transactions
- Designed and authored DBS outbound and inbound integration connectors to facilitate Fast and PayNow payments and ICN notifications

Habitto Money GraphQL and REST backend

- Lead consultant for setting up initial backend monorepo and related pipelines
- Setup initial Nx monorepo, BitBucket OpenID runner authentication, and inital Terraform setup
- Later, consultant and lead developer for authoring fully automated self provisioning review environments using BitBucket CI/CD with multiple deployables
- Lead developer for authoring a seamless custom in-house replacement solution for Fivetran

Dealt Australia

- Lead architect for migrating existing customer-hosted Hyperledger Fabric to ImmuDB
- Automated all application deployments and configuration management using AWS CodeDeploy and BitBucket CI/CD
- Migrated all existing infrastructure to use IaC for management and provisioning and added enforcement checks to prevent manual infastructure manipulations
- Revamped virtual cloud networking to incorporate public, private, and NAT subnets and redesigned the microservice entrypoints and inter-service communication
- Standardized logging, monitoring, and alerting

CaterCare Australia

- Lead consultant and developer for standardizing GitHub developer workflows
- Wrote a custom extensible GitHub Action to facilitate auto-labelling and auto-approving for a specific set of pull-requests obeying a predefined ruleset that require no reviews
- Set up branch/tag protection rules, code owners, secret scanning, and repo properties to standardize merging and pull-request approvals

honors and awards

HackerRank
Gold Honours - Problem Solving

Gold honours, five-star proficiency rating, in the domains of algorithms and data structures.

HackerRank
Gold Honours - Python

Gold honours, five-star proficiency rating, in the domain of Python subsuming all language-supported paradigms.

HackerRank
Gold Honours - SQL

Gold honours, five-star proficiency rating, in the domains of SQL subsuming all language-supported paradigms.

licenses & certifications

Databricks
Platform Administrator

Databricks
Azure Databricks Platform Architect

Databricks
AWS Databricks Platform Architect

HashiCorp
IaC Automation

HackerRank
Python

academic service

freeCodeCamp
Curriculum Contributor February 2017 - Present

Authore guides and specimen solutions for freeCodeCamp certification program challenges on algorithm design.

edX
Content Support Associate December 2016 - Present

Validate and alter qualitative attributes of uploaded course materials to meet edX guidelines.

community service

United Nations
Volunteer Task Force January 2015 - Present